

75 AWS Interview Questions for DevOps

IAM, EKS, VPC & More — Basics to Advanced

A structured progression through AWS — from IAM and networking to EKS, disaster recovery, and cost optimization.

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Level 1: Core AWS & IAM (Q1-18)

Q1. What is AWS IAM and why is it central to AWS security?

IAM (Identity and Access Management) controls who (users, groups, roles) can do what (actions) on which resources in an AWS account. It's central to security because nearly every AWS security incident traces back to overly broad or misconfigured permissions — IAM is the primary control plane for access.

Q2. What is the difference between an IAM user, an IAM group, and an IAM role?

An IAM user represents a specific person or application with long-term credentials. A group is a collection of users sharing the same permissions, for easier management. A role is an identity with temporary credentials that can be assumed by users, applications, or AWS services — preferred over long-lived user credentials for most workloads.

Q3. What is the difference between an IAM policy and a resource-based policy?

An IAM (identity-based) policy is attached to a user, group, or role and defines what that identity can do across AWS. A resource-based policy is attached directly to a resource (like an S3 bucket policy or an SQS queue policy) and defines who can access that specific resource, evaluated together with identity-based policies.

Q4. What is the principle of least privilege, and how do you enforce it in practice?

Least privilege means granting only the exact permissions needed to perform a task, nothing more. In practice: use scoped IAM policies instead of wildcard actions/resources, prefer roles with temporary credentials over long-lived access keys, use IAM Access Analyzer to audit unused permissions, and apply permission boundaries for delegated administration.

Q5. What is an IAM policy evaluation logic — how does AWS decide allow vs deny when multiple policies apply?

An explicit Deny in any applicable policy always wins, overriding any Allow. If there's no explicit Deny, AWS checks for an explicit Allow across all applicable identity-based, resource-based, and permission boundary policies. If neither applies, the default is implicit Deny.

Q6. What is the difference between an IAM role and an EC2 instance profile?

An instance profile is a container that allows an IAM role to be passed to and used by an EC2 instance — you can't directly attach an IAM role to an EC2 instance without wrapping it in an instance profile, though the AWS console does this automatically when you attach a role via the UI.

Q7. What is AWS STS and what role does it play with IAM roles?

STS (Security Token Service) issues temporary, short-lived security credentials when a role is assumed (via AssumeRole or similar calls), rather than relying on long-term static credentials. This is the mechanism underlying cross-account access, federated login, and workload identity for services like EC2 and Lambda.

Q8. What is the difference between a Security Group and a Network ACL?

A Security Group is a stateful, instance-level firewall — return traffic is automatically allowed regardless of outbound rules. A Network ACL is a stateless, subnet-level firewall where you must explicitly allow both inbound and outbound traffic, evaluated in numbered rule order, with an explicit deny option that Security Groups don't have.

Q9. What is the AWS shared responsibility model?

AWS is responsible for security 'of' the cloud — physical data centers, hardware, and the underlying virtualization/networking infrastructure. The customer is responsible for security 'in' the cloud — data classification, IAM configuration, guest OS patching (for EC2), network configuration, and application-level security.

Q10. What is an AWS Organization and what is it used for?

AWS Organizations lets you centrally manage multiple AWS accounts under one umbrella — consolidating billing, applying Service Control Policies (SCPs) to restrict what member accounts can do regardless of their own IAM policies, and organizing accounts into Organizational Units for governance at scale.

Q11. What is a Service Control Policy (SCP) and how does it differ from an IAM policy?

An SCP is applied at the AWS Organizations level and sets the maximum available permissions for accounts within an OU — it's a guardrail, not a grant, meaning it can only restrict what's possible, never actually grant permissions itself. IAM policies, by contrast, are what actually grant permissions to identities within the boundaries SCPs allow.

Q12. What is Amazon S3 and what are its core storage classes?

S3 is AWS's object storage service, storing data as objects within buckets. Core storage classes include S3 Standard (frequent access), S3 Infrequent Access (cheaper, retrieval fee), S3 Glacier (archival, slow retrieval, very cheap), and S3 Intelligent-Tiering (automatically moves objects between tiers based on access patterns).

Q13. What is the difference between an S3 bucket policy and IAM policies for controlling S3 access?

An IAM policy is attached to a user/role and defines what that identity can do across AWS, including S3. A bucket policy is attached directly to the bucket and defines who (which principals, potentially including cross-account) can access it — useful for cross-account sharing and public access control, and evaluated together with any applicable IAM policies.

Q14. What is Amazon RDS and how does Multi-AZ deployment improve availability?

RDS is a managed relational database service (MySQL, PostgreSQL, etc.) that handles backups, patching, and failover. Multi-AZ deployment maintains a synchronously replicated standby instance in a different Availability Zone, and automatically fails over to it if the primary becomes unavailable — improving availability without requiring manual intervention.

Q15. What is the difference between an Application Load Balancer (ALB) and a Network Load Balancer (NLB)?

ALB operates at Layer 7, routing based on HTTP path/host headers and ideal for microservices and content-based routing. NLB operates at Layer 4, handling raw TCP/UDP traffic with ultra-low latency and static IPs, ideal for extreme performance needs or non-HTTP protocols.

Q16. What is an Auto Scaling Group (ASG) and how does it work?

An ASG maintains a desired number of EC2 instances, automatically launching new instances (from a launch template) when scaling out based on metrics like CPU or request count, and terminating instances when scaling in — keeping capacity aligned with demand without manual intervention.

Q17. What is the difference between a public subnet and a private subnet in a VPC?

A public subnet has a route to an Internet Gateway, allowing resources within it to be directly reachable from (and reach out to) the internet. A private subnet has no direct route to an Internet Gateway — outbound internet access, if needed, typically goes through a NAT Gateway in a public subnet, while inbound access from the internet is blocked entirely.

Q18. What is CloudTrail and why is it important for security and compliance?

CloudTrail logs every API call made within an AWS account — who made it, when, from where, and what action was taken. It's essential for security investigations (tracing what happened during an incident), compliance auditing, and detecting unauthorized or unexpected activity across the account.

Level 2: VPC & Networking (Q19-33)

Q19. What is a VPC and what are its core components?

A VPC (Virtual Private Cloud) is an isolated, logically separated virtual network within AWS. Core components include subnets (public/private), route tables (define traffic routing rules), an Internet Gateway (enables internet access), and Security Groups/NACLs (traffic filtering).

Q20. What is a NAT Gateway and why is it needed?

A NAT Gateway allows resources in a private subnet to initiate outbound connections to the internet (e.g., to download packages or call an external API) while preventing unsolicited inbound connections from the internet — private resources stay unreachable from outside, but can still reach out when needed.

Q21. What is the difference between a NAT Gateway and a NAT Instance?

A NAT Gateway is a fully managed AWS service — highly available within its AZ, automatically scaled, no patching required. A NAT Instance is a manually managed EC2 instance configured to perform NAT — cheaper for very low-traffic use cases but requires you to manage its availability, scaling, and patching yourself.

Q22. What is VPC Peering and what is its main limitation at scale?

VPC Peering creates a direct network connection between two VPCs, allowing resources in each to communicate as if on the same network. Its main limitation is that it's non-transitive — if VPC A is peered with B, and B is peered with C, A cannot reach C through B — making full-mesh peering complex to manage as the number of VPCs grows.

Q23. What is AWS Transit Gateway and how does it solve the VPC Peering scaling problem?

Transit Gateway acts as a central hub that VPCs (and on-prem networks via VPN/Direct Connect) connect to, rather than requiring a full mesh of individual peering connections between every pair. This dramatically simplifies network topology and management as the number of VPCs grows into the dozens or hundreds.

Q24. What is the difference between a route table and a Security Group?

A route table determines where network traffic is directed at the subnet level (e.g., 'traffic to 0.0.0.0/0 goes to the Internet Gateway'). A Security Group is a firewall that determines whether traffic is allowed to reach a specific resource at all, evaluated independently of and after routing decisions.

Q25. What is a VPC Endpoint and why would you use one?

A VPC Endpoint allows private connectivity between your VPC and supported AWS services (like S3 or DynamoDB) without traffic leaving the AWS network or requiring a NAT Gateway/Internet Gateway — improving security (no internet exposure) and often reducing data transfer costs.

Q26. What is the difference between a Gateway VPC Endpoint and an Interface VPC Endpoint?

A Gateway Endpoint (supported only for S3 and DynamoDB) works by adding a route table entry, at no additional cost. An Interface Endpoint uses an Elastic Network Interface with a private IP within your subnet (powered by AWS PrivateLink), supports a much broader range of services, but incurs an hourly cost.

Q27. How would you troubleshoot an EC2 instance in a private subnet that can't reach the internet?

Verify the private subnet's route table has a route to a NAT Gateway (not an Internet Gateway directly), confirm the NAT Gateway itself is healthy and sits in a public subnet with its own valid route to an Internet Gateway, check the instance's Security Group and the subnet's NACL allow the outbound traffic, and confirm DNS resolution/hostnames are enabled on the VPC if relying on DNS.

Q28. What is the difference between an Elastic IP and a standard public IP assigned to an EC2 instance?

A standard public IP is dynamically assigned and changes if the instance is stopped and started again. An Elastic IP is a static, persistent public IP you allocate to your account and can attach/detach/re-attach to instances as needed, remaining constant even through stop/start cycles.

Q29. What is the difference between AWS Direct Connect and a Site-to-Site VPN?

Direct Connect is a dedicated, private physical network connection from your on-premises data center to AWS, offering consistent low latency and higher bandwidth at a higher cost and longer setup time. Site-to-Site VPN establishes an encrypted tunnel over the public internet — faster and cheaper to set up, but subject to internet variability in latency and throughput.

Q30. How does DNS resolution work for resources within a VPC?

AWS provides a default DNS resolver (the Amazon-provided DNS server, typically at the VPC's base +2 address) that resolves both public DNS names and internal, VPC-specific hostnames for resources like EC2 instances and RDS endpoints — this can be customized via VPC DHCP option sets if a custom DNS server is needed instead.

Q31. What is a Network Load Balancer's 'static IP' feature and why might that matter for certain use cases?

An NLB provides a fixed, static IP address per Availability Zone (or lets you bring your own Elastic IP), which matters for use cases like firewall allowlisting by clients that need a predictable IP to permit — a capability an ALB, which uses dynamically resolved DNS-based IPs, doesn't offer.

Q32. What is the difference between subnetting a VPC by Availability Zone versus by application tier, and why does AZ-based design matter for high availability?

Spreading subnets (and the resources within them) across multiple Availability Zones ensures that the failure of a single AZ's underlying data center doesn't take down your entire application, since AZs are physically isolated. Subnetting purely by tier (web/app/db) without also spanning multiple AZs would leave you fully exposed to a single AZ failure.

Q33. How would you design network security for a three-tier application (web, app, database) in AWS?

Place the web tier in public subnets (behind an ALB) with a Security Group only allowing HTTP/HTTPS from the internet, the app tier in private subnets with a Security Group only allowing traffic from the web tier's Security Group, and the database tier in a separate private subnet with a Security Group only allowing traffic from the app tier — layering access so each tier only accepts traffic from the tier immediately in front of it.

Level 3: EKS & Containers (Q34-50)

Q34. What is Amazon EKS and how does it differ from self-managed Kubernetes on EC2?

EKS is a managed Kubernetes service where AWS operates and scales the control plane (API server, etcd) for high availability across multiple AZs. Self-managed Kubernetes on EC2 requires you to provision, patch, secure, and scale every control plane component yourself — EKS removes that operational burden at the cost of a per-cluster hourly control plane fee.

Q35. What is the difference between EKS managed node groups and self-managed (unmanaged) worker nodes?

Managed node groups let AWS handle provisioning, lifecycle management (updates, scaling, graceful draining during upgrades) of the underlying EC2 instances via Auto Scaling Groups. Self-managed nodes give you full control over the AMI, configuration, and lifecycle, but require you to handle all of that automation yourself.

Q36. What is Fargate and how does it change the operational model for running containers on EKS?

Fargate is a serverless compute engine that runs Pods without you provisioning or managing any underlying EC2 instances at all — AWS handles the compute layer entirely. This removes node-level patching, capacity planning, and scaling concerns, at the trade-off of less control over the underlying host and typically higher cost per unit of compute than well-utilized EC2.

Q37. How does IAM authentication work for accessing an EKS cluster?

EKS uses the aws-auth ConfigMap (or, in newer versions, EKS Access Entries) to map specific IAM users/roles to Kubernetes RBAC users/groups — meaning an IAM identity alone doesn't grant Kubernetes-level permissions; it must be explicitly mapped to a Kubernetes role or group to actually perform actions within the cluster.

Q38. What is IRSA (IAM Roles for Service Accounts) and what problem does it solve?

IRSA lets you associate a specific IAM role with a Kubernetes service account, so Pods using that service account can assume fine-grained AWS permissions scoped just to their needs, using an OIDC identity provider tied to the cluster. It solves the problem of previously having to grant broad IAM permissions to entire worker nodes (and therefore every Pod on them), replacing that with precise, Pod-level AWS access control.

Q39. What is the AWS Load Balancer Controller and what does it do for an EKS cluster?

It's a Kubernetes controller that watches Ingress and Service resources and automatically provisions and configures native AWS Application Load Balancers or Network Load Balancers to match, letting you manage AWS load balancing declaratively through standard Kubernetes manifests rather than manual AWS console configuration.

Q40. What is the EKS control plane's networking relationship to your VPC?

The EKS control plane runs in an AWS-managed VPC, but is configured with elastic network interfaces (ENIs) into your VPC's subnets so it can communicate with your worker nodes — meaning proper subnet and security group configuration in your own VPC is still essential for the control plane to reach your nodes and Pods correctly.

Q41. What is the Amazon VPC CNI plugin and how does Pod networking work on EKS by default?

The VPC CNI plugin assigns each Pod a real IP address directly from the VPC's IP address range (rather than using an overlay network), making Pods first-class citizens on the VPC network. This gives excellent network performance but means Pod IP capacity is constrained by the ENI/IP limits of the underlying EC2 instance type, which is a common scaling consideration.

Q42. How do you handle running out of available IP addresses for Pods in an EKS cluster using the VPC CNI?

Options include using larger instance types (which support more ENIs/IPs), enabling prefix delegation (which assigns a block of IPs per ENI rather than one at a time, dramatically increasing Pod density per node), using larger or additional subnet CIDR ranges, or switching to a secondary, non-routable CIDR range for Pods specifically.

Q43. What is the EKS Cluster Autoscaler and how does it interact with Auto Scaling Groups?

Cluster Autoscaler monitors for Pods that can't be scheduled due to insufficient node capacity and, when found, increases the desired count of the relevant Auto Scaling Group to add nodes; it also scales down underutilized nodes when their Pods could be safely rescheduled elsewhere, working alongside (not instead of) the ASG's own mechanics.

Q44. How would you secure secrets used by applications running on EKS?

Avoid plain Kubernetes Secrets for highly sensitive data (they're only base64-encoded, not encrypted, by default), and instead use AWS Secrets Manager or Systems Manager Parameter Store combined with the AWS Secrets and Configuration Provider (ASCP) for the Kubernetes Secrets Store CSI Driver — or enable envelope encryption of Kubernetes Secrets at the etcd level using AWS KMS.

Q45. What is the difference between using ECR (Elastic Container Registry) versus a public registry like Docker Hub for an EKS deployment?

ECR is a private, AWS-native container registry, tightly integrated with IAM for access control and with built-in vulnerability scanning, and avoids Docker Hub's rate limiting for pulls. Using ECR keeps images within your AWS security boundary and IAM permission model rather than relying on an external registry's availability and access controls.

Q46. How do you perform a safe EKS cluster version upgrade?

Upgrade the control plane first (EKS handles this with minimal disruption since it's managed and highly available), then upgrade managed node groups (which handles cordoning and draining nodes gracefully) or your self-managed node infrastructure, checking add-on compatibility (VPC CNI, CoreDNS, kube-proxy) and application compatibility with the new Kubernetes API version at each step, following AWS's documented supported upgrade path one minor version at a time.

Q47. What is EKS Pod Identity and how does it compare to IRSA?

EKS Pod Identity is a newer, simplified mechanism for granting Pods AWS permissions, removing the need to manually configure an OIDC provider and complex trust policies that IRSA requires — it's generally easier to set up and manage while achieving the same fine-grained, Pod-level IAM permission goal.

Q48. How would you monitor and troubleshoot networking issues between Pods across different worker nodes in EKS?

Verify Security Groups attached to the worker nodes/ENIs allow the necessary Pod-to-Pod traffic (the VPC CNI means Pod traffic is subject to the underlying node's Security Group rules), check VPC Flow Logs to see if traffic is being dropped at the network level, and use standard Kubernetes networking debugging (checking Service endpoints, NetworkPolicies) alongside AWS-specific network layer checks.

Q49. What cost optimization strategies are specific to running EKS at scale?

Use Spot Instances for fault-tolerant workloads via managed node groups with mixed instance policies, right-size node instance types based on actual Pod resource requests rather than over-provisioning, leverage Fargate selectively for spiky or infrequent workloads where paying per-Pod is cheaper than maintaining idle EC2 capacity, and use Cluster Autoscaler aggressively to avoid paying for consistently underutilized nodes.

Q50. Describe a real production issue you've encountered running workloads on EKS or another AWS-managed service, and what you learned from it.

A strong answer names the specific failure (e.g., running out of Pod IPs under a traffic spike, an IAM misconfiguration blocking IRSA, a node group failing to scale during an incident), the diagnostic path taken, the fix, and a concrete change made afterward (enabling prefix delegation, adjusting IAM trust policies, tuning Cluster Autoscaler) so the same issue can't silently recur.

Level 4: Advanced & Cost/Reliability Scenarios (Q51-75)

Q51. How would you design a highly available, multi-AZ architecture on AWS for a production web application?

Deploy across at least two or three Availability Zones with an ALB distributing traffic, an Auto Scaling Group spanning those AZs for the compute tier, a Multi-AZ RDS instance (or read replicas across AZs) for the database layer, and S3/CloudFront for static assets — eliminating single points of failure at every tier of the architecture.

Q52. What is AWS CloudWatch and what role does it play in a DevOps workflow?

CloudWatch collects metrics, logs, and events from AWS resources and applications, and supports alarms that trigger notifications or automated actions (like Auto Scaling) based on thresholds. It's the primary native observability tool for monitoring AWS infrastructure health and application behavior.

Q53. What is the difference between a CloudWatch Alarm and an EventBridge rule?

A CloudWatch Alarm watches a specific metric against a threshold and triggers an action (notification, Auto Scaling) when breached — focused on numeric metric monitoring. EventBridge routes events (state changes, API calls, scheduled triggers) from a much broader range of sources to targets like Lambda functions or Step Functions — a general-purpose event bus rather than metric-threshold monitoring.

Q54. How would you design a disaster recovery strategy across AWS regions?

Approaches range in cost and recovery time: backup and restore (cheapest, slowest recovery — data backed up cross-region, infrastructure rebuilt from IaC when needed), pilot light (minimal always-on infrastructure in the DR region, scaled up during failover), warm standby (a scaled-down but fully functional copy always running), to multi-region active-active (most expensive, near-zero recovery time, full production capacity in both regions simultaneously).

Q55. What is the difference between RTO and RPO, and why do they matter for DR planning?

RTO (Recovery Time Objective) is the maximum acceptable time to restore service after a failure. RPO (Recovery Point Objective) is the maximum acceptable amount of data loss, measured in time (e.g., 'we can lose at most 15 minutes of data'). Together they drive which DR strategy is appropriate and cost-justified for a given system's business criticality.

Q56. How does AWS Key Management Service (KMS) work, and where would you use it?

KMS creates and manages cryptographic keys used to encrypt data across AWS services (S3, EBS, RDS, Secrets Manager) and in custom applications via the KMS API. It supports fine-grained IAM-based access control over who can use or manage specific keys, and integrates with CloudTrail for auditing key usage.

Q57. What is the difference between server-side encryption with an AWS-managed key (SSE-S3) versus a customer-managed KMS key (SSE-KMS) for S3?

SSE-S3 uses keys fully managed by AWS with no visibility or control for the customer over key policies or rotation. SSE-KMS uses a customer-managed KMS key, giving you control over key policies, audit trails of key usage via CloudTrail, and the ability to revoke or rotate the key independently — at the cost of KMS API request pricing.

Q58. How would you troubleshoot an application experiencing intermittent 5xx errors behind an ALB?

Check the ALB's target group health checks to see if targets are being marked unhealthy and cycling in/out, review CloudWatch metrics for the ALB (HTTPCode_ELB_5XX vs HTTPCode_Target_5XX to distinguish load-balancer-level vs application-level errors), check application logs and resource utilization on the backend instances/Pods, and verify connection draining/deregistration delay settings aren't causing requests to be dropped during scaling events.

Q59. What is AWS Systems Manager Session Manager and why might you prefer it over SSH for accessing EC2 instances?

Session Manager provides secure shell access to EC2 instances through the AWS API without requiring an open inbound SSH port, a bastion host, or managing SSH key pairs — access is controlled entirely via IAM permissions and every session is logged, improving both the security posture and auditability compared to traditional SSH access.

Q60. How would you approach cost optimization for a large, established AWS environment?

Use Cost Explorer and Trusted Advisor to identify idle or underutilized resources (unattached EBS volumes, idle load balancers, oversized instances), right-size EC2/RDS instances based on actual CloudWatch utilization data, use Reserved Instances or Savings Plans for predictable steady-state workloads, use Spot Instances for fault-tolerant workloads, and implement S3 lifecycle policies to transition infrequently accessed data to cheaper storage tiers automatically.

Q61. What is the difference between a Reserved Instance and a Savings Plan?

A Reserved Instance commits to a specific instance type and region/AZ for a 1 or 3-year term in exchange for a significant discount, but is relatively inflexible if your instance needs change. A Savings Plan commits to a certain dollar amount of compute usage per hour, offering similar discounts but with much more flexibility to change instance families, sizes, and even switch between EC2, Fargate, and Lambda while still benefiting from the discount.

Q62. How would you detect and respond to a potential security breach in an AWS account (e.g., an exposed access key)?

Immediately deactivate/rotate the exposed credential in IAM, use CloudTrail to review exactly what actions were taken using that credential during the exposure window, use GuardDuty (if enabled) for automated threat detection insight into anomalous activity, and follow an incident response runbook to assess and remediate any resources that may have been created or modified maliciously.

Q63. What is AWS GuardDuty and what kind of threats does it detect?

GuardDuty is a threat detection service that continuously analyzes CloudTrail logs, VPC Flow Logs, and DNS logs using machine learning and threat intelligence feeds to identify suspicious activity — such as API calls from known malicious IPs, unusual instance behavior suggesting compromise (like cryptocurrency mining), or reconnaissance activity indicating an account may be under attack.

Q64. How would you design an AWS environment to support strict compliance requirements (e.g., HIPAA, PCI-DSS)?

Use AWS Organizations with SCPs to enforce guardrails across all accounts, enable CloudTrail and AWS Config across the organization for full auditability and configuration compliance tracking, encrypt data at rest (KMS) and in transit (TLS) everywhere required, restrict network access using private subnets and VPC endpoints to minimize internet exposure, and use AWS Config rules or a third-party tool to continuously monitor for compliance drift.

Q65. What is the difference between AWS Config and CloudTrail?

CloudTrail records a log of API calls and actions taken (who did what, when). AWS Config tracks the configuration state of resources over time and can evaluate that configuration against defined compliance rules — CloudTrail tells you what happened, Config tells you what your resources currently look like and whether they comply with policy.

Q66. How would you handle a scenario where an EC2-based application needs zero-downtime deployments without using Kubernetes?

Use an Auto Scaling Group behind an ALB with a rolling or blue-green deployment strategy (via CodeDeploy, which natively supports both), ensuring health checks gate new instances from receiving traffic until they're verified healthy, and that old instances are only terminated after new ones are confirmed healthy and serving traffic successfully.

Q67. What is the significance of AWS's per-service API rate limits, and how do you design around them in automation/pipelines?

AWS enforces rate limits (throttling) on API calls to prevent any single account or caller from overwhelming a service; hitting these limits returns throttling errors that automation must handle gracefully. Design around this with exponential backoff and retry logic (the AWS SDKs implement this by default), batching API calls where possible, and being mindful of tight polling loops in scripts or Terraform providers that could trigger throttling at scale.

Q68. How would you architect a serverless CI/CD pipeline using native AWS services?

CodeCommit or GitHub as the source, CodeBuild for running build/test stages, CodeDeploy for the actual deployment (supporting blue-green/canary for EC2, ECS, and Lambda), and CodePipeline orchestrating the stages end-to-end — with CloudWatch Events/EventBridge triggering the pipeline on source changes and SNS for pipeline notifications.

Q69. What is AWS Well-Architected Framework and why might an interviewer ask about it?

It's AWS's set of best-practice pillars (Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability) for designing cloud architectures. Interviewers ask about it to gauge whether you think about trade-offs holistically — not just 'does it work,' but whether it's secure, cost-efficient, and reliable by design.

Q70. Describe a real incident involving AWS infrastructure that you've resolved, and what changed afterward.

A strong answer names the specific failure (an Auto Scaling Group failing to scale during a traffic spike, a misconfigured Security Group blocking legitimate traffic, an accidental public S3 bucket), the diagnostic and remediation steps taken under pressure, and a concrete, lasting change made afterward (a CloudWatch alarm added, an SCP guardrail implemented, a runbook written) so the same failure mode is caught earlier or prevented entirely next time.

Q71. What is the difference between horizontal and vertical scaling on AWS, and which do most cloud-native architectures favor?

Vertical scaling means increasing the resources (CPU/memory) of a single instance. Horizontal scaling means adding more instances of the same size. Most cloud-native AWS architectures favor horizontal scaling since it works naturally with Auto Scaling Groups and load balancers, avoids a hard ceiling on a single instance's maximum size, and improves fault tolerance by spreading load across multiple instances.

Q72. What is AWS Lambda and what workloads is it well-suited (and poorly suited) for?

Lambda is a serverless compute service that runs code in response to events without provisioning or managing servers, billed per invocation and execution duration. It's well-suited for short-lived, event-driven, bursty workloads (API backends, file processing triggers); it's poorly suited for long-running processes (execution

time limits), workloads needing consistent low-latency 'warm' starts at all times, or very high-throughput steady-state compute where EC2/containers are more cost-effective.

Q73. How would you handle secrets rotation for a database credential used by multiple applications across AWS?

Use AWS Secrets Manager's automatic rotation feature, which can rotate the credential on a schedule and update it in place without requiring you to redeploy every consuming application — applications fetch the current credential dynamically from Secrets Manager at runtime rather than having it hardcoded or baked into configuration, so rotation is transparent to them.

Q74. What is the difference between ECS and EKS, and when would you choose one over the other?

ECS is AWS's own proprietary container orchestration service — simpler to learn and operate, tightly integrated with other AWS services, but locked into AWS. EKS is managed Kubernetes — a portable, industry-standard API with a much larger ecosystem of tools, but a steeper learning curve and more operational complexity. Teams already invested in Kubernetes expertise or needing multi-cloud portability typically choose EKS; teams wanting AWS-native simplicity often choose ECS.

Q75. How would you approach migrating a legacy on-premises application to AWS with minimal downtime?

Common patterns include the 'strangler fig' approach (gradually routing traffic to newly migrated components while the legacy system still handles the rest), setting up hybrid connectivity (Direct Connect/VPN) during the transition period, replicating data continuously to AWS ahead of cutover, and performing the final cutover during a low-traffic window with a well-rehearsed rollback plan in case issues arise post-migration.